ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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PHYSICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

EM Waves & Wave Optics

English Medium

EXERCISE (O-1)

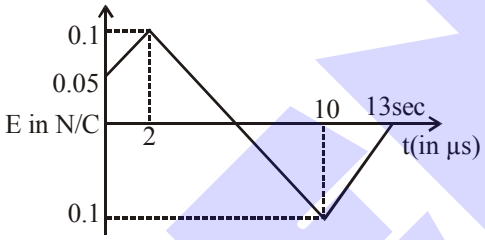
A. EM Waves

1. Instantaneous displacement current of 1.0A in the space between the parallel plates of 1μF capacitor can be established by changing potential difference of:

(A) 10^{-6} V/s (B) 10^{6} V/s (C) 10^{-8} V/s (D) 10^{8} V/s

WO0083

2. The electric field through an area of $2m^2$ varies with time as shown in the graph. The greatest displacement current through the area is at :-



(A) t = 1 sec. (B) t = 4 sec. (C) t = 8 sec. (D) t = 12 sec.

WO0089

3. A plane electromagnertic wave travels in free space along x-axis. At a particular point in space, the electric field along y-axis is 9.3 Vm^{-1}. The magnetic induction is:

(A) 3.1×10^{-8} T (B) 3×10^{-5} T (C) 3.1×10^{-6} T (D) 9.3×10^{-6} T

WO0088

4. A plane electromagnetic wave travelling along the X-direction has a wavelength of 3mm. The variation in the electric field occurs in the Y-direction with an amplitude 66Vm^{-1}.The equation for the electric and magnetic fields as a function of x and t are respectively.

(A) $E_y = 33 \cos \pi \times 10^{11} \left(t - \frac{x}{c}\right)$; $B_z = 1.1 \times 10^{-7} \cos \pi \times 10^{11} \left(t - \frac{x}{c}\right)$

(B) $E_y = 11 \cos 2\pi \times 10^{11} \left(t - \frac{x}{c}\right)$; $B_y = 1.1 \times 10^{-7} \cos 2\pi \times 10^{11} \left(t - \frac{x}{c}\right)$

(C) $E_x = 33 \cos \pi \times 10^{11} \left(t - \frac{x}{c}\right)$; $B_x = 1.1 \times 10^{-7} \cos \pi \times 10^{11} \left(t - \frac{x}{c}\right)$

(D) $E_y = 66 \cos 2\pi \times 10^{11} \left(t - \frac{x}{c}\right)$; $B_z = 2.2 \times 10^{-7} \cos 2\pi \times 10^{11} \left(t - \frac{x}{c}\right)$

WO0090

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5. The electric field of a plane polarized electromagnetic wave in free space at time t = 0 is given by an expression $\vec{E}(x,y) = 10\hat{j}\cos[(6x + 8z)]$ The magnetic field $\vec{B}$ (x, z, t) is given by: (c is the velocity of light)

(A) $\frac{1}{c}\left(6\hat{k}+8\hat{i}\right)\cos\left[\left(6x-8z+10ct\right)\right]$ (B) $\frac{1}{c}\left(6\hat{k}-8\hat{i}\right)\cos\left[\left(6x+8z-10ct\right)\right]$

(C) $\frac{1}{c}\left(6\hat{k}+8\hat{i}\right)\cos\left[\left(6x+8z-10ct\right)\right]$ (D) $\frac{1}{c}\left(6\hat{k}-8\hat{i}\right)\cos\left[\left(6x+8z+10ct\right)\right]$

WO0196

6. A plane electromagnetic wave of frequency 500 MHz is travelling in vacuum along y-direction. At a particular point in space and time, $\vec{B} = 8.0\times10^{-8}\,\hat{z}T$. The value of electric field at this point is : (speed of light = 3×10^8 ms^{-1}) $\hat{x}, \hat{y}, \hat{z}$ are unit vectors along x, y and z direction.

[JEE Mains_2021]

(A) $-24\hat{x}$ V / m (B) $2.6\hat{x}$ V / m (C) $24\hat{x}$ V / m (D) $-2.6\hat{y}$ V / m

WO0198

7. Microwave oven acts on the principle of transferring electrons from lower to
(A) higher energy levels in water molecule
(B) giving rotational energy to water molecules
(C) giving translational energy to water molecules
(D) giving vibrational energy to water molecules

WO0199

8. If the wavelength of an electromagnetic radiation is of the order of size of a football field, which radiation is it ?
(A) Microwave (B) Gamma ray
(C) Radio wave (D) x-ray

WO0200

9. The amplitude of electric field in a parallel light beam of intensity 4Wm^{-2} is :
(A) 35.5 NC^{-1} (B) 45.5 NC^{-1} (C) 49.5 NC^{-1} (D) 54.8 NC^{-1}

WO0082

10. A plane electromagnetic wave,
$E_z = 100\cos(6\times10^8 t + 4x)$ V/m
propagates in a medium of dielectric constant:
(A) 1.5 (B) 2.0 (C) 2.4 (D) 4.0

WO0084

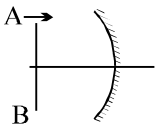
11. The rms value of the electric field of the light coming from the sun is 720 N/c. The average total energy density of the electromagnetic wave is :
(A) 4.58×10^{-6} J/m^3 (B) 6.37×10^{-9} J/m^3
(C) 81.35×10^{-12} J/m^3 (D) 3.3×10^{-3} J/m^3

WO0086

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B. Young's double slit experiment and it's modification

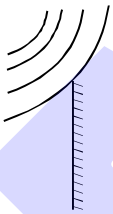
12. A plane wavefront AB is incident on a concave mirror as shown. Then, the wavefront just after reflection is :

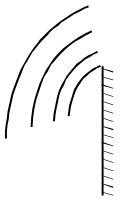
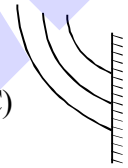
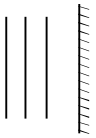


(A)  (B)  (C)  (D) None of the above

WO0301

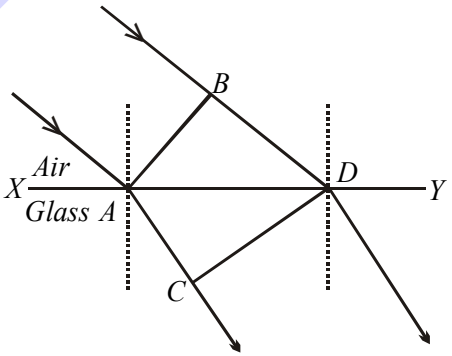
13. Spherical wave fronts shown in figure, strike a plane mirror. Reflected wave fronts will be as shown in



(A)  (B)  (C)  (D) 

WO0302

14. A wave front AB moving in air is incident on a plane glass surface XY. CD is the refracted wave front. The refractive index of glass with respect to air is



(A) $\frac{BD}{AC}$ (B) $\frac{AB}{CD}$ (C) $\frac{BD}{AD}$ (D) $\frac{AC}{AD}$

WO0303

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15. An interference pattern is obtained by Young's double slit arrangement and the fringe width is β. If the distance between the slits is halved and the distance of the screen from the slits is doubled, the new fringe width will be

(A) 0.25β (B) 2β (C) 4β (D) 8β

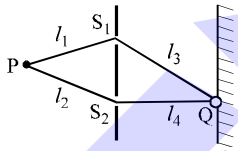
WO0304

16. Two slits separated by a distance of 1 mm are illuminated with light of wave length $6.5\times10^{-7}m$. The interference fringes are observed on a screen placed 1m from the slits. The distance between the third dark fringe and the fifth bright fringe is equal to

(A) 0.65 mm (B) 1.63 mm (C) 3.25 mm (D) 4.88 mm

WO0305

17. Two identical narrow slits S_1 and S_2 are illuminated by light of wavelength λ from a point source P. If, as shown in the diagram, the light is then allowed to fall on a screen, and if n is a positive integer, the condition for destructive interference at Q is :-



(A) $(\ell_1 - \ell_2) = (2n + 1)\lambda/2$ (B) $(\ell_3 - \ell_4) = (2n + 1)\lambda/2$

(C) $(\ell_1 + \ell_2) - (\ell_3 + \ell_4) = n\lambda$ (D) $(\ell_1 + \ell_3) - (\ell_2 + \ell_4) = (2n + 1)\lambda/2$

WO0036

18. In YDSE, the source placed symmetrically with respect to the slit is now moved parallel to the plane of the slits so that it is closer to the upper slit, as shown. Then,



(A) the fringe width will increase and fringe pattern will shift down.
(B) the fringe width will remain same but fringe pattern will shift up.
(C) the fringe width will decrease and fringe pattern will shift down.
(D) the fringe width will remain same but fringe pattern will shift down.

WO0045

19. In Young's experiment, using red and blue lights of wavelengths 7800Å and 5200Å respectively, the value of n for which nth red fringe coincides with (n + 1)th blue fringe is

(A) 2 (B) 3 (C) 4 (D) 5

WO0306

20. In Young's double-slit experiment, we get 60 fringes in the field of view if we use light of wavelength 4000Å. The number of fringes we will get in the same field of view if we use light of wavelength 6000Å is

(A) 40 (B) 90 (C) 60 (D) 50

WO0307

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21. Two monochromatic and coherent point sources of light are placed at a certain distance from each other in the horizontal plane. The locus of all those points in the horizontal plane which have constructive interference will be

(A) a hyperbola
(B) family of hyperbolas
(C) family of straight lines
(D) family of parabolas

WO0058

22. In the interference pattern, energy is

(A) created at the positions of maxima
(B) destroyed at the positions of minima
(C) conserved but is redistributed
(D) not conserved

WO0308

23. Two beams of light having intensities I and $4I$ interfere to produce a fringe pattern on a screen. The phase difference between the beams is $\pi/2$ at point A and π at point B. Then the difference between the resultant intensities at A and B is : **[IIT-JEE (Scr.) 2001]**

(A) 2I
(B) 4I
(C) 5I
(D) 7I

WO0028

24. Two coherent monochromatic light beams of intensities I and $4I$ are superposed. The maximum and minimum possible intensities in the resulting beam are :

(A) $5I$ and I
(B) 5I and 3I
(C) $9I$ and I
(D) $9I$ and $3I$

WO0030

25. In a Young's double slit experiment, a small detector measures an intensity of illumination of I units at the centre of the fringe pattern. If one of the two (identical) slits is now covered, the measured intensity will be :-

(A) 2I
(B) I
(C) I/4
(D) I/2

WO0038

26. In a double slit experiment, the separation between the slits is $d = 0.25$ cm and the distance of the screen D = 100 cm from the slits. If the wavelength of light used is $\lambda = 6000$Å and I_0 is the intensity of the central bright fringe, the intensity at a distance $x = 4 \times 10^{-5}$ m from the central maxima is :-

(A) I_0
(B) $I_0/2$
(C) $3I_0/4$
(D) $I_0/3$

WO0056

27. In Young's double slit experiment, one of the slit is wider than other, so that the amplitude of the light from one slit is double of that from other slit. If I_m be the maximum intensity, the resultant intensity I when they interfere at phase difference ϕ is given by : **[AIEEE-2012]**

(A) $\frac{I_m}{9}(1 + 8\cos^2\frac{\phi}{2})$
(B) $\frac{I_m}{9}(4 + 5\cos\phi)$
(C) $\frac{I_m}{3}(1 + 2\cos^2\frac{\phi}{2})$
(D) $\frac{I_m}{5}(1 + 4\cos^2\frac{\phi}{2})$

WO0117

28. In Young's double slit arrangement, water is filled in the space between screen and slits. Then :

(A) fringe pattern shifts upwards but fringe width remains unchanged.
(B) fringe width decreases and central bright fringe shifts upwards.
(C) fringe width increases and central bright fringe does not shift.
(D) fringe width decreases and central bright fringe does not shift.

WO0043

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29. Light of wavelength λ in air enters a medium of refractive index μ. Two points in this medium, lying along the path of this light, are at a distance x apart. The phase difference between these points is :

(A) $\frac{2\pi\mu x}{\lambda}$ (B) $\frac{2\pi x}{\mu\lambda}$ (C) $\frac{2\pi(\mu-1)x}{\lambda}$ (D) $\frac{2\pi x}{(\mu-1)\lambda}$

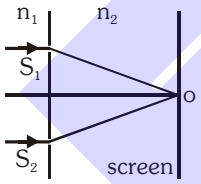
WO0044

30. Interference fringes were produced in Young's double-slit experiment using light of wavelength 5000Å. When a film of thickness $2.5\times10^{-3}\,cm$ was placed in front of one of the slits, the fringe pattern shifted by a distance equal to 20 fringewidths. The refractive index of the material of the film is

(A) 1.25 (B) 1.356 (C) 1.4 (D) 1.5

WO0309

31. In a Young's double slit experiment the slits S_1 & S_2 are illuminated by a parallel beam of light of wavelength 4000 Å, from the medium of refractive index $n_1 = 1.2$. A thin film of thickness 1.2 μm and refractive index n = 1.5 is placed infront of S_1 perpendicular to path of light. The refractive index of medium between plane of slits & screen is $n_2 = 1.4$. If the light coming from the film and S_1 & S_2 have equal intensities I then intensity at geometrical centre of the screen O is



(A) 0 (B) 2I (C) 4I (D) None of these

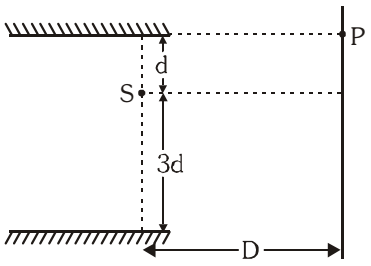
WO0310

32. A parallel beam of light of wavelength 560 nm falls on a thin film of oil in air (refractive index = 1.4). What should be the minimum thickness of the film so that it weakly transmits the light?

(A) 100 nm (B) 200 nm (C) 300 nm (D) 400 nm

WO0311

33. Consider the optical system shown in figure. The point source of light S is having wavelength equals to λ.The light is reaching screen only after reflection. For point P to be 2nd maxima, the value of λ would be (D >> d and d >> λ)



(A) $\frac{12d^2}{D}$ (B) $\frac{6d^2}{D}$ (C) $\frac{3d^2}{D}$ (D) $\frac{24d^2}{D}$

WO0312

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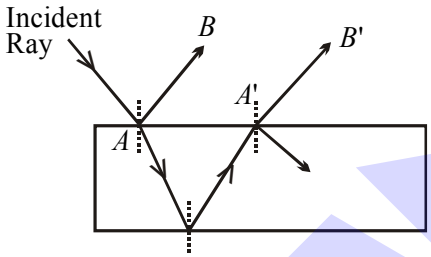
34. When viewed in white light, soap bubbles show colours because of
(A) interference (B) scattering (C) diffraction (D) dispersion

WO0313

35. Oil floating on water appears coloured due to interference of light. The approximate thickness of oil for such effect to be visible is
(A) 1 cm (B) 1 mm (C) 10,000Å (D) 100Å

WO0314

36. A ray of light is incident on a parallel sided glass slab at a point A as shown in the fig., and undergoes partial reflection and refraction. At each reflection 25%, of the incident energy is reflected. The rays AB and $A'B'$ interfere. The ratio of maximum and minimum intensities in the interference is



(A) 7 : 1 (B) 4 : 1 (C) 49 : 1 (D) 16 : 1

WO0315

37. Light of wavelength 540 nm is falling on wedge of glass (refractive index 1.5). Interference takes place between the rays reflected from top and bottom faces of the wedge. The wedge increases thickness from one dark fringe to the next by
(A) 90 nm (B) 180 nm (C) 280 nm (D) 360 nm

WO0316

C. Diffraction

38. Light of wavelength 6000 Å is incident normally on a slit of width 24×10^{-5} cm. Find out the angular position of second minimum from central maximum–
(A) 15° (B) 30° (C) 60° (D) 74°

WO0317

39. To observe diffraction, the size of the obstacle
(A) should be of the same order as the wavelength
(B) should be much larger than the wavelength
(C) has no relation to wavelength
(D) should be exactly half the wavelength

WO0318

40. Diffraction pattern of single slit consists of a central band which is
(A) wider, brighter and accompanied with alternate dark and bright bands of decreasing intensity
(B) narrow, bright with alternate dark and bright bands of equal intensity
(C) wider, bright with alternate dark and bright bands of equal intensity
(D) dark and alternate bright bands of equal intensity

WO0319

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41. A slit of width a is illuminated by red light of wavelength 6500Å. The first minimum will fall at $\theta = 30°$ if a is

(A) 3250Å (B) 6.5×10^{-4} *mm* (C) 1.3 micron (D) 2.6×10^{-4} *cm*

WO0320

42. Consider Fraunhofer diffraction pattern obtained with a single slit illuminated at normal incidence. At the angular position of the first diffraction minimum, the phase difference (in radians) between the wavelets from the opposite edges of the slit is

(A) π (B) 2π (C) $\pi/4$ (D) $\pi/2$

WO0321

43. In an electron microscope, if the accelerating voltage is increased from 20 kV to 80 kV, the resolving power of the microscope will become

(A) R/4 (B) 4R (C) 2R (D) R/2

WO0322

44. Two points separated by a distance of 0.1 mm can just be inspected in microscope when light of wavelength 6000 A° is used. If the light of wavelength 4800 A° is used, this limit of resolution will become.

(A) 0.80 mm (B) 0.12 mm (C) 0.10 mm (D) 0.08 mm

WO0323

45. The resolution limit of the eye is $1'$. At a distance of x km from the eye, two persons stand with a lateral separation of 3 metres. For the two persons to be just resolved by the naked eye, x should be

(A) 10 km (B) 15 km (C) 20 km (D) 30 km

WO0324

46. In a single slit diffraction pattern, as the width of the slit is increased,

(A) the peak intensity of central maxima increases and its width also increases

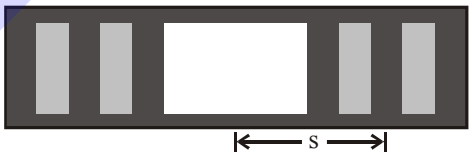
(B) the peak intensity of central maxima increases and its width decreases.

(C) the peak intensity of central maxima decreases and its width increases

(D) the peak intensity of central maxima decreases and its width also decreases.

WO0097

47. Light of wavelength 600 nm is incident upon a single slit with width 4×10^{-4} m. The figure shows the pattern observed on a screen positioned 2 m from the slits. Determine the distance s.



(A) 0.002 m (B) 0.003 m (C) 0.004 m (D) 0.006 m

WO0095

D. Polarization

48. According to electromagnetic theory of light, the polarisation of electromagnetic waves is determined by

(A) the electric field only (B) the magnetic field only

(C) both electric and magnetic fields (D) direction of propagation

WO0325

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49. When light is incident on a transparent surface at the polarizing angle, which of the following is completely polarized ?

(A) Reflected light

(B) Refracted light

(C) Both reflected as well as refracted light

(D) Neither reflected nor refracted light

WO0326

50. Plane polarised light is passed through a polaroid. In viewing through the polaroid, we find that when the polaroid is given one complete rotation about the direction of light

(A) the intensity of light gradually decreases to zero and remains at zero

(B) the intensity of light gradually increases to a maximum and remains maximum

(C) there is no change in the intensity of light

(D) the intensity of light varies such that it is twice maximum and twice zero

WO0327

51. The axes of the polariser and analyser are inclined to each other at 45°. If the amplitude of the unpolarised light incident on the polariser is a, then what is the amplitude of the light transmitted through the analyser ?

(A) $a/2$ (B) $a/\sqrt{2}$ (C) $\frac{\sqrt{3}}{2}a$ (D) $\frac{3}{4}a$

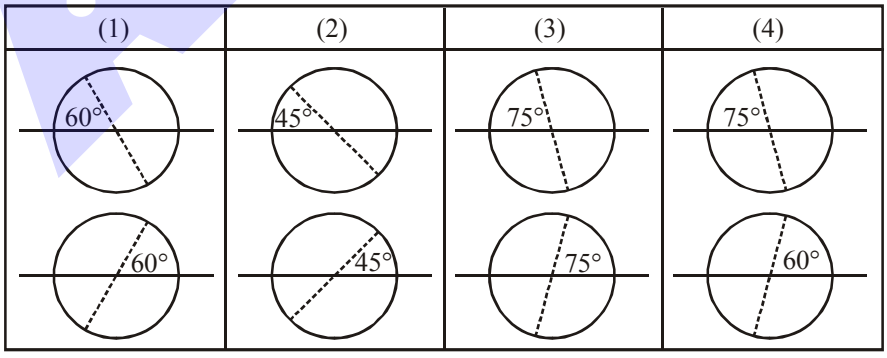
WO0328

52. Two polaroid sheets are placed one over the other with their axes inclined to each other at 30°. The ratio of the intensity of the unpolarised incident light and the polarised emergent light is

(A) 1/2 (B) 3/4 (C) 8/3 (D) 3/1

WO0329

53. The diagrams show four pairs of polarizing sheets, with the polarizing directions indicated by dashed lines. The two 'sheets of each pair are placed one behind the other and the front sheet is illuminated by unpolarized light. The incident intensity is the same for all pairs of sheets. Rank the pairs according to the intensity of the transmitted light, least to greatest.



(A) 1,2, 3, 4 (B) 4, 2, 1, 3 (C) 2, 4, 3, 1 (D) 2, 1,4, 3

WO0094

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54. A parallel beam of monochromatic and unpolarised light is incident on a transparent slab of refractive index $\sqrt{3}$. The reflected beam is completely polarised ? What are the angles of incidence (i) and refraction (r) ?

(A) $i = 30°,\ r = 30°$ (B) $i = 60°,\ r = 60°$ (C) $i = 30°,\ r = 60°$ (D) $i = 60°,\ r = 30°$

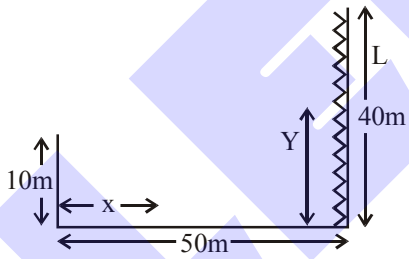
WO0330

55. A beam of light strikes a piece of glass at an angle of incidence of 60° and the reflected beam is completely plane polarised. The refractive index of the glass is :-

(A) 1.5 (B) $\sqrt{3}$ (C) $\sqrt{2}$ (D)(3/2)

WO0092

56. A person lives in a high-rise building on the bank of a river 50m wide. Across the river is a well lit tower of height 40 m. When the person, who is at a height of 10m, looks through a polarizer at an appropriate angle at light of the tower reflecting from the river surface, he notes that intensity of light coming from distance X from his building is the least and this corresponds to the light coming from light bulbs at height 'Y' on the tower. The values of X and Y are respectively close to (refractive index of water $\simeq$ 4/3)



(A) 13 m, 27 m (B) 22 m, 13 m (C) 25 m, 10 m (D) 17 m, 29 m

WO0099

ALLEN®

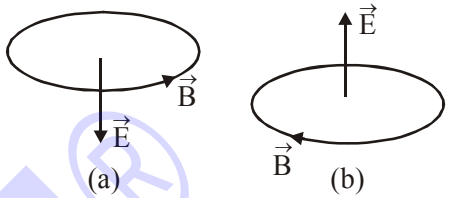
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound\Wave-EM\Wave & WaveOp\Eng\03_EM-Wave & WO\01_Th+Ex.p65

EXERCISE (O-2)

A. EM Waves

1. Figure shows, in two situations and electric field vector $\vec{E}$ and an induced magnetic field line. In each, is the magnitude of $\vec{E}$ increasing or decreasing?

(A) increasing in a and decreasing in b.
(B) increasing in a and b.
(C) decreasing in a and b.
(D) decreasing in a and increasing in b.



WO0194

2. A conducting circular loop made of a thin wire, has area 3.5×10^{-3} m^2 and resistance l0Ω. It is placed perpendicular to a time dependent magnetic field B(t) = (0.4T)sin(50πt). The field is uniform in space. Then the net charge flowing through the loop during t = 0 s and t = 10 ms is close to :

(A) 0.14mC (B) 0.21 mC (C) 0.06 mC (D) 0.07 mC

WO0195

3. A plane electromagnetic wave having a frequency ν = 23.9 GHz propagates along the positive z-direction in free space. The peak value of the electric field is 60 V/m. Which among the following is the acceptable magnetic field component in the electromagnetic wave ?

(A) $\vec{B} = 2\times10^{7}\sin(0.5\times10^{3}z + 1.5\times10^{11}t)\hat{i}$ (B) $\vec{B} = 2\times10^{-7}\sin(1.5\times10^{2}x + 0.5\times10^{11}t)\hat{j}$

(C) $\vec{B} = 2\times10^{-7}\sin(0.5\times10^{3}z - 1.5\times10^{11}t)\hat{i}$ (D) $\vec{B} = 60\sin(0.5\times10^{3}x + 1.5\times10^{11}t)\hat{k}$

WO0197

4. An electromagnetic wave in vacuum has the electric and magnetic fields $\vec{E}$ and $\vec{B}$, which are always perpendicular to each other. The direction of polarization is given by $\vec{X}$ and that ofwave propagation by $\vec{k}$. Then **[AIEEE-2012]**

(A) $\vec{X} \parallel \vec{B}$ and $\vec{k} \parallel \vec{E}\times\vec{B}$ (B) $\vec{X} \parallel \vec{E}$ and $\vec{k} \parallel \vec{B}\times\vec{E}$

(C) $\vec{X} \parallel \vec{B}$ and $\vec{k} \parallel \vec{B}\times\vec{E}$ (D) $\vec{X} \parallel \vec{E}$ and $\vec{k} \parallel \vec{E}\times\vec{B}$

WO0118

B. Young's double slit experiment and it's modification

5. In a YDSE, the central bright fringe can be identified :

(A) as it has greater intensity than the other bright fringes.
(B) as it is wider than the other bright fringes.
(C) as it is narrower than the other bright fringes.
(D) by using white light instead of single wavelength light.

WO0033

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6. In Young's double slit experiment, the two slits act as coherent sources of equal amplitude A and wavelength λ. In another experiment with the same setup the two slits are sources of equal amplitude A and wavelength λ but are incoherent. The ratio of the average intensity of light at the midpoint of the screen in the first case to that in the second case is :-

(A) 1 : 1 (B) 2 : 1 (C) 4 : 1 (D) none of these

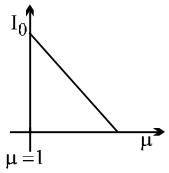
WO0037

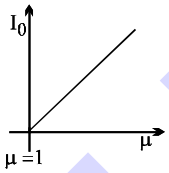
7. In a Young's double slit experiment D equals the distance of screen and d is the separation between the slit. The distance of the nearest point to the central maximum where the intensity is same as that due to a single slit, is equal to :-

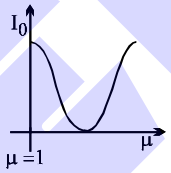
(A) $\frac{D\lambda}{d}$ (B) $\frac{D\lambda}{2d}$ (C) $\frac{D\lambda}{3d}$ (D) $\frac{2D\lambda}{d}$

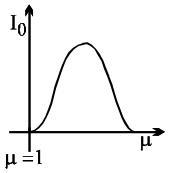
WO0039

8. In a YDSE experiment if a slab whose refractive index can be varied is placed in front of one of the slits then the variation of resultant intensity at mid-point of screen with μ ($\mu \geq 1$)will be best represented by [Assume slits of equal width and there is no absorption by slab]

(A) 

(B) 

(C) 

(D) 

WO0047

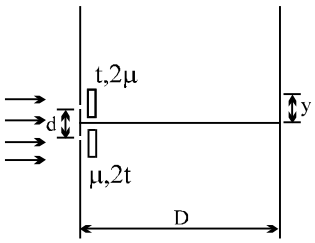
9. In a Young's double-slit experiment, let A and B be the two slits. Films of thicknesses t_A and t_B and refractive indices μ_A and μ_B, are placed in front of A and B respectively. If $\mu_A t_A = \mu_B t_B$, the central maximum will :

(A) not shift (B) shift towards A

(C) shift towards B (D) option (B), if $t_B > t_A$; option (C) if $t_B < t_A$

WO0048

10. In the YDSE shown the two slits are covered with thin sheets having thickness t & $2t$ and refractive index 2μ and μ. Find the position (y) of central maxima



(A) zero (B) $\frac{tD}{d}$ (C) $-\frac{tD}{d}$ (D) None

WO0049

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11. In a double slit experiment, when the width of one slit is made twice as wide as the other in compared to normal YDSE having slits of equal width. Then, in the interference pattern **[IIT-JEE(Scr.) 2000]**
(A) the intensities of both the maxima and the minima increase.
(B) the intensity of the maxima increases and the minima has zero intensity.
(C) the intensity of the maxima decreases and that of the minima increases.
(D) the intensity of the maxima decreases and the minima has zero intensity.

WO0050

12. **Statement-1 :** In standard YDSE set up with visible light, the position on screen where phase difference is zero appears bright.
and
Statement-2 : In YDSE set up magnitude of electromagnetic field at central bright fringe is not varying with time.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

WO0054

13. **Statement-1 :**In YDSE, the spacing between any two successive points having intensity half of the maximum intensity is same.
and
Statement-2 : The intensity on the screen in YDSE varies uniformly with distance from central maximum.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

WO0055

14. Imagine a Young's double slit interference experiment performed with waves associated with fast moving electrons produced from an electron gun. The distance between successive maxima will decrease the most if :-
(A) the accelerating voltage in the electron gun is decreased
(B) the accelerating voltage is increased and the distance of the screen from the slits is decreased
(C) the distance of the screen from the slits is increased.
(D) the distance between the slits is decreased.

WO0057

C. Diffraction

15. Light with wavelength 500 nm falls on a slit of width 10 μm at an angle 30° to its normal. Find the angular position of the first minima located on both sides of the central Fraunhofer maximum :-
(A) 33° and 27° (B) 25° and 35° (C) 36° and 24° (D) 28° and 32°

WO0205

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16. The image of a star (effectively a point source) is made by a convergent lens of focal length 1m and diameter of aperture 5.0 cm. If the lens is ideal and the effective wavelength in image formation is taken as 5×10^{-5} cm, the diameter of the image formed will be nearest to:

(A) zero (B) 10^{-6} cm (C) 10^{-5} cm (D) 10^{-3} cm

WO0096

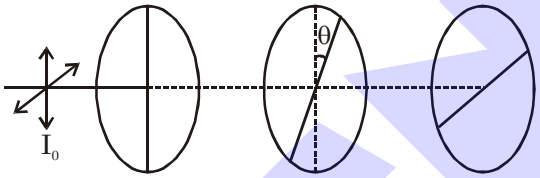
17. A microscope objective gathers light over a cone of semi-vertex 30° and used visible light of wavelength 5500 Å. Its resolving limit is :-

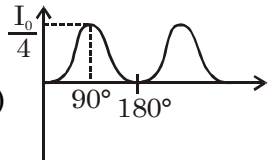
(A) 2.5×10^{-5} cm (B) 4.6×10^{-5} cm (C) 3.9×10^{-5} cm (D) 6.7×10^{-5} cm

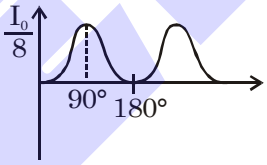
WO0206

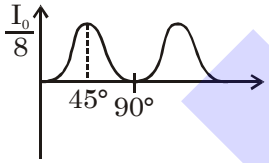
D. Polarization

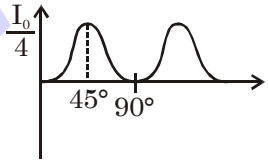
18. Two polarizing filters are crossed at 90°. A third filter is placed between them initially aligned with first. It is then rotated by angle θ. Find plot of intensity coming from third and final filter as a function of θ. The incident light is unpolarized with intensity I_0 :-



(A) 

(B) 

(C) 

(D) 

WO0202

19. For a partially polarized light, we can define the degree of polarization. The degree of polarization of light is given by $P = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}$. Here I_{max} and I_{min} are maximum and minimum possible intensities in two mutually perpendicular directions. For a given partially polarized light beam, the value of P is 0.25. The approximate ratio of maximum and minimum amplitudes in two mutually perpendicular directions is:

(A) 4 (B) 1.66 (C) 1.3 (D) 9

WO0203

20. The Sun is directly overhead and you are facing toward the north. Light coming to your eyes from the sky just above the horizon is :-

(A) Partially polarized north-south (B) Partially polarized east,west

(C) Partially polarized up-down (D) Randomly polarized

WO0093

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21. **Statement-1:** On viewing the clear blue portion of the sky through a Calcite Crystal, the intensity of transmitted light varies as the crystal is rotated.

Statement-1: The light coming from the sky is polarized due to scattering of sun light by particles in the atmosphere. The scattering is largest for blue light. **[AIEEE-2011]**

(A) Statement-1 is false, statement-2 is true

(B) Statement-1 is true, statement-2 is false

(C) Statement-1 is true, statement-2 true; statement-2 is the correct explanation of statement-1

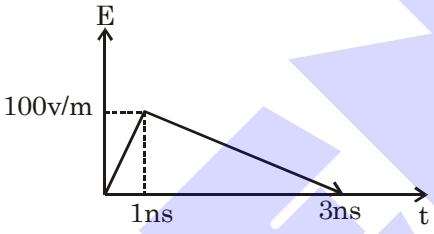
(D) Statement-1 is true, statement-2 is true; statement -2 is not correct explanation of statement-1.

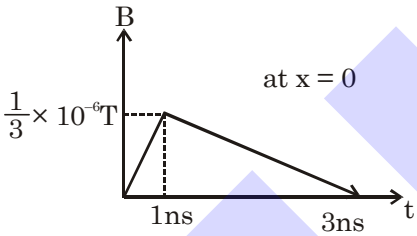
WO0114

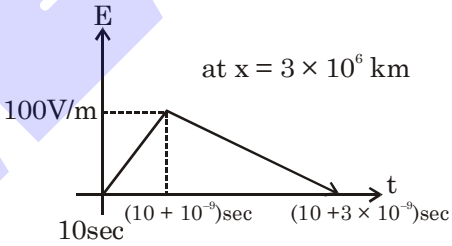
MULTIPLE CORRECT TYPE QUESTIONS

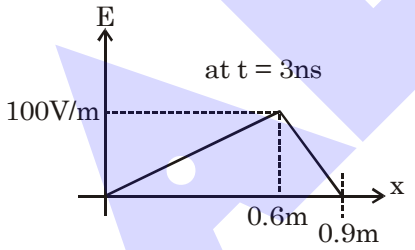
A. EM Waves

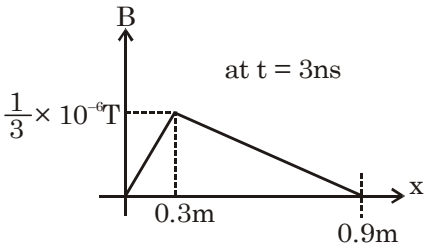
22. Consider a pulse of plane electromagnetic wave whose electric field is in y direction and is propagating in +x direction. The electric field at x = 0 varies as shown in the graph. Choose the correct graph (s) for electric field and magnetic field as x and t.



(A)


(B)


(C)


(D)


WO0207

23. A silver wire has resistivity of 1.62×10^{-8} Ωm and a cross-sectional area of 5.00 mm^2. The current in the wire is uniform and changing at the rate of 2000 A/s when the current is 100 A.

(A) the magnitude of the (uniform) electric field in the wire when the current in the wire is 100 A is 0.324 V/m.

(B) displacement current in the wire at that time 2.87×10^{-16} A.

(C) the ratio of the magnitude of the magnetic field due to the displacement current to that due to the current at a distance r from the wire is 2.87×10^{-18}.

(D) The displacement current density lines are circular.

WO0208

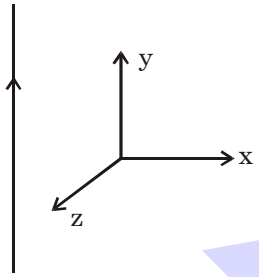
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24. A capacitor with parallel circular plates of radius R = 1.20 cm is discharging via a current of 12.0 A. The maximum induced magnetic field has a magnitude of 12.0 mT. At what radius (b) inside and (c) outside the capacitor gap is the magnitude of the induced magnetic field 3.00 mT?

(A) 0.3 cm (B) 2.7 cm (C) 4.8 cm (D) 5.4 cm

WO0209

25. Consider an infinite wire carrying current $i = \frac{i_0 t}{T}$, Here i_0 & T are constant, t is time. The wire is in xy plane as shown. At the origin,



(A) Displacement current is in +ve y direction.

(B) Electric field is in +ve z direction.

(C) Poynting vector $\frac{(\vec{E}\times\vec{B})}{\mu_0}$ is in +ve x direction.

(D) Magnetic field is in –ve z direction.

WO0210

26. The Poynting vector S of an electromagnetic wave is given by

$$\vec{S} = 942 W/m^2 \cos^2\left(10x - 3\times10^9 t\right)\hat{i}$$

where x is in meters and t is in seconds.

(A) The electric field can be approx 600 N/C $\cos\left(10x - 3\times10^9 t\right)\hat{j}$

(B) The magnetic field can be approx 2×10^{-6} T $\cos\left(10x - 3\times10^9 t\right)\hat{i}$

(C) The wave is propagating in positive x direction

(D) The wave is propagating in negative x direction

WO0211

B. Young's double slit experiment and it's modification

27. In the Young's double-slit experiment, the interference pattern is found to have an intensity ratio between bright and dark fringes as 9. This implies that

(A) the intensities at the screen due to the two slits are 5 units and 4 units respectively

(B) the intensities at the screen due to the two slits are 4 units and 1 unit respectively

(C) the amplitude ratio is 6

(D) the amplitude ratio is 2

WO0331

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28. To observe a sustained interference pattern formed by two light waves, it is not necessary that they must have :

(A) the same frequency
(B) same amplitude
(C) a constant phase difference
(D) the same intensity

WO0059

29. As a wave propagates, **[IIT-JEE 1999]**

(A) the wave intensity remains constant for a plane wave
(B) the wave intensity decreases with distance from source and is proportional to inverse of the distance, for a spherical wave.
(C) the wave intensity decreases with distance from source and proportional to inverse square of the distance, for a spherical wave.
(D) total average power of the spherical wave over any spherical surface centered at the source remains same.

WO0060

30. In a YDSE apparatus, if we use white light then :

(A) the fringe next to the central will be red
(B) the central fringe will be white.
(C) the fringe next to the central will be violet
(D) there will not be a completely dark fringe.

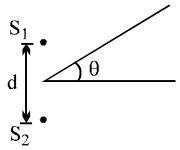
WO0061

31. In a Young's double-slit experiment, let A and B be the two slits. A thin film of thickness t and refractive index μ is placed in front of A. Let b = fringe width. The central maximum will shift :

(A) towards A (B) towards B (C) by $t(\mu - 1)\dfrac{\beta}{\lambda}$ (D) by $\mu t \dfrac{\beta}{\lambda}$

WO0063

32. In an interference arrangement similar to Young's double- slit experiment, the slits S_1 & S_2 are illuminated with coherent microwave sources, each of frequency 10^6 Hz. The sources are synchronized to have zero phase difference. The slits are separated by a distance d = 150.0 m. The intensity $I(\theta)$ is measured as a function of θ at a large distance from S_1 & S_2, where θ is defined as shown. If I_0 is the maximum intensity then $I(\theta)$ for $0 \le \theta \le 90^\circ$ is given by :



(A) $I(\theta) = \dfrac{I_0}{2}$ for $\theta = 30^\circ$
(B) $I(\theta) = \dfrac{I_0}{4}$ for $\theta = 90^\circ$
(C) $I(\theta) = I_0$ for $\theta = 0^\circ$
(D) $I(\theta)$ is constant for all values of θ .

WO0065

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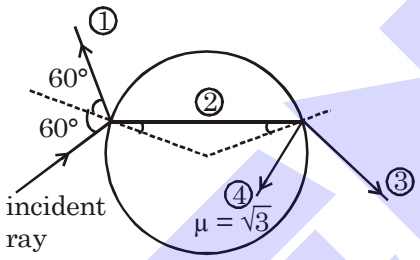
33. In a YDSE with two identical slits, when the upper slit is covered with a thin, perfectly transparent sheet of mica, the intensity at the centre of screen reduces to 75% of the initial value. Second minima is observed to be above this point and third maxima below it. Which of the following can be a possible value of phase difference caused by the mica sheet

(A) $\frac{\pi}{3}$ (B) $\frac{13\pi}{3}$ (C) $\frac{17\pi}{3}$ (D) $\frac{11\pi}{3}$

WO0067

D. Polarization

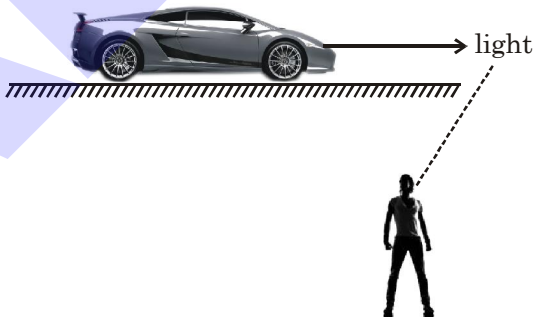
34. Consider an unpolarized light ray incident on a transparent sphere of refractive index $\sqrt{3}$ as shown. Which of the following statement (s) are correct ?



(A) Ray 1 is completely polarized. (B) Ray 2 is partially polarized.
(C) Ray 4 is completely polarized. (D) Ray 3 is partially polarized.

WO0212

35. Light coming from an automobile headlights is normally polarized such that plane of electric field is vertical. This light strikes dust particles in air & is scattered. Suppose an observer is standing ⊥ to road as shown. The light that he receives is only scattered light. The light is

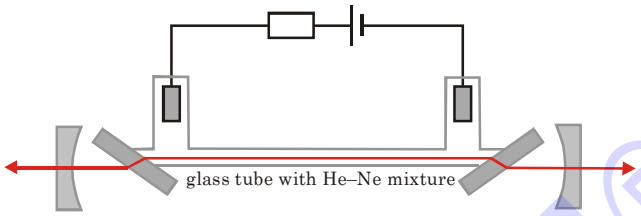


(A) Having no intensity in direction of observer.
(B) Has electric field in vertical plane ⊥ to car's velocity.
(C) Has magnetic field parallel to car's velocity but in horizontal plane.
(D) Has magnetic field in vertical plane parallel to car's velocity.

WO0213

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36. A Brewster window is used to minimize losses by reflection. A typical example is a helium-neon laser with a sealed glass tube as shown in Figure, where glass windows separate the laser gas mixture from ambient air. The light coming from laser is polarized. This light emanating from the glass tube strikes the two glass plates at each of the ends as shown. If the conditions are appropriate, there is no partial reflection from the glass plates. The light (shown by red colour) is then focused using convex lens shown at each of the ends. The refractive index of glass can be assumed to be 1.5 and that of both the sides of glass can be assumed to be 1.



The appropriate condition(s) mentioned in the write-up is (are)

(A) The angle made by each of the glass plate with the direction of light beam should be the same

(B) The angle between the light beam and the glass plate surface should be $\tan^{-1}(2/3)$

(C) The incident beam should be plane polarized with the plane of electric field same as the plane of paper

(D) The incident beam should be plane polarized with the plane of electric field perpendicular to the plane of paper

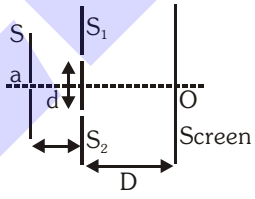
WO0214

COMPREHENSION TYPE QUESTIONS

B. Young's double slit experiment and it's modification

Paragraph for Question No. 37 to 39

The figure shows a schematic diagram showing the arrangement of Young's Double Slit Experiment



37. Choose the correct statement(s) related to the wavelength of light used

(A) Larger the wavelength of light larger the fringe width

(B) The position of central maxima depends on the wavelength of light used

(C) If white light is used in YDSE, then the violet colour forms its first maxima closest to the central maxima

(D) The central maxima of all the wavelengths coincide

WO0070

38. If the distance *D* is varied, then choose the correct statement(s)

(A) The angular fringe width does not change

(B) The fringe width changes in direct proportion

(C) The change in fringe width is same for all wavelengths

(D) The position of central maxima remains unchanged

WO0070

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39. If the distance d is varied, then identify the correct statement(s)

(A) The angular width does not change

(B) The fringe width changes in inverse proportion

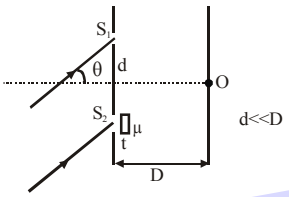
(C) The positions of all maxima change

(D) The positions of all minima change

WO0070

Paragraph for Question No. 40 to 42

A monochromatic beam of light falls on Young's double slit experiment apparatus as shown in figure. A thin sheet of glass is inserted in front of lower slit S_2.



40. The central bright fringe can be obtained

(A) at O only (B) at O or below O only

(C) at O or above O only (D) Anywhere on the screen

WO0071

41. If central bright fringe is obtained on screen at O :-

(A) $(\mu-1)\,t = d\sin\theta$ (B) $(\mu-1)\,t = d\cos\theta$

(C) $(\mu-1)t + d\sin\theta = 0$ (D) $\dfrac{t}{\mu-1} = \dfrac{d}{\sin\theta}$

WO0071

42. The phase difference between the waves interfering at fifth minima is

(A) 5π (B) 7π (C) 9π (D) 11π

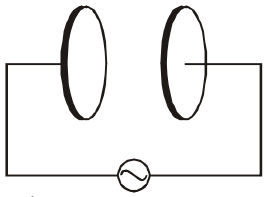
WO0071

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EXERCISE (S)

A. EM Waves

1. A parallel plate capacitor (figure) made of circular plates each of radius R = 6.0 cm has a capacitance C = 100 μF. The capacitor is connected to a 230 V ac supply with a (angular) frequency of 300 rad s^{-1}.



(a) What is the rms value of the conduction current?
(b) Is the conduction current equal to the displacement current?
(c) Determine the amplitude of B at a point 3.0 cm from the axis between the plates.

WO0072

2. A parallel-plate capacitor is formed by two discs with a uniform poorly conducting medium between them. The capacitor was initially charged and then disconnected from a voltage source. Neglecting the edge effects, show that there is no magnetic field between capacitor plates.

WO0166

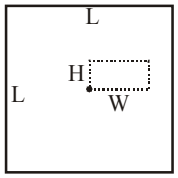
3. A parallel-plate air condenser whose each plate has an area S = 100 cm^2 is connected in series to an ac circuit. Find the electric field strength amplitude in the capacitor if the sinusoidal current amplitude in lead wires is equal to I_m = 1.0 mA and the current frequency equals ω = 1.6 × 10^7 s^{-1}.

WO0167

4. A long straight solenoid has n turns per unit length. An alternating current I = I_m sin ωt flows through it. Find the displacement current density as a function of the distance r from the solenoid axis. The cross-sectional radius of the solenoid equals R.

WO0168

5. A capacitor with square plates of edge length L is being discharged by a current of 0.75A. Figure is a head-on view of one of the plates from inside the capacitor. A dashed rectangular path is shown. If L = 12 cm, W = 4.0 cm, and H = 2.0 cm, what is the value of $\int \vec{B}\cdot d\vec{s}$ around the dashed path?



WO0169

6. Suppose that the electric field part of an electromagnetic wave in vaccum is E = {(3.1 N/C) cos [(1.8 rad/m) y + (5.4 × 10^8 rad/s)t] $\hat{i}$.
(a) What is the direction of propagation?
(b) What is the wavelength λ?
(c) What is the frequency v?
(d) What is the amplitude of the magnetic field part of the wave?
(e) Write an expression for the magnetic field part of the wave.

WO0075

7. In a plane electromagnetic wave, the electric field oscillates sinusoidally at a frequency of 2.0 × 10^{10}Hz and amplitude 48 V m^{-1}.
(a) What is the wavelength of the wave?
(b) What is the amplitude of the oscillating magnetic field?
(c) Show that the average energy density of the E field equals the average energy density of the B field. [c = 3 × 10^8 m s^{-1}]

WO0074

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8. An electromagnetic wave of frequency 5 GHz, is travelling in a medium whose relative electric permittivity and relative magnetic permeability both are 2. Its velocity in this medium is _____ × 10^7 m/s. **[JEE Mains_2021]**

WO0173

9. A current flowing in the winding of a long straight solenoid is increased at a sufficiently slow rate. Demonstrate that the rate at which the energy of the magnetic field in the solenoid increases is equal to the flux of the Poynting vector across the lateral surface of the solenoid.

WO0174

10. At one instant, the electric and magnetic fields at one point of an electromagnetic wave arc $\vec{E} = (200\hat{i} + 300\hat{j} - 50\hat{k})\,V/m$ and $\vec{B} = B_0(7.3\hat{i} - 7.3\hat{j} + a\hat{k})\,\mu T$

a. Whal are the values of a and B_0 ?

b. What is the Poynting vector at this time and position?

WO0176

11. Find an expression for the electric field of a plane electromagnetic wave with the following properties: (a) the frequency is 10^{14} Hz: (b) the wave travels in a medium of index of refraction 1.4; (c) the wave propagates along a line that lies in the xy-plane and makes a 30° angle with the x-axis; (d) the wave is polarized along the z-axis; (e) the average value of the Poynting vector is 500 W/m^2.

WO0177

B. Young's double slit experiment and it's modification

12. Two coherent waves are described by the following expressions.

$$E_1 = E_0 \sin\left(\frac{2\pi x_1}{\lambda} - 2\pi ft + \frac{\pi}{6}\right);\ E_2 = E_0 \sin\left(\frac{2\pi x_2}{\lambda} - 2\pi ft + \frac{\pi}{8}\right)$$

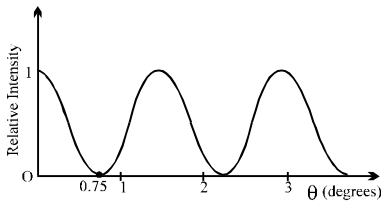
Determine the relationship between x_1 and x_2 that produces constructive interference when the two waves are superposed.

WO0078

13. In Young's double slit experiment the slits are 0.5 mm apart and the interference is observed on a screen at a distance of 100 cm from the slit. It is found that the 9th bright fringe is at a distance of 7.5mm from the second dark fringe from the centre of the fringe pattern on same side. Find the wavelength of the light used.

WO0002

14. Light of wavelength 520 nm passing through a double slit, produces interference pattern of relative intensity versus angular position θ as shown in the figure. Find the separation d between the slits.



WO0003

15. The distance between two slits in a YDSE apparatus is 3mm. The distance of the screen from the slits is 1m. Microwaves of wavelength 1 mm are incident on the plane of the slits normally. Find the distance of the first maxima on the screen from the central maxima. Also find the total number of maxima on the screen.

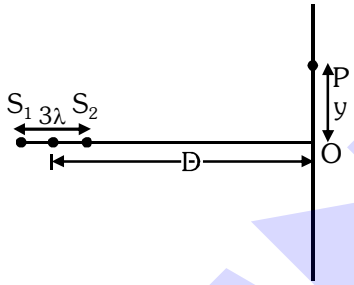
WO0006

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16. One slit of a double slit experiment is covered by a thin glass plate of refractive index 1.4 and the other by a thin glass plate of refractive index 1.7. The point on the screen, where central bright fringe was formed before the introduction of the glass sheets, is now occupied by the 5th bright fringe. Assuming that both the glass plates have same thickness and wavelength of light used is 4800 Å, find their thickness.

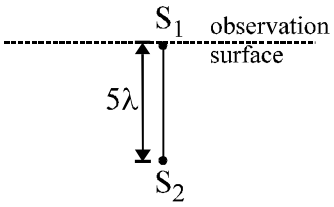
WO0007

17. Figure shows two coherent microwave source S_1 and S_2 emitting waves of wavelength λ and separated by a distance 3λ. For $\lambda << D$ and $y \neq 0$, the minimum value of y for point P to be an intensity maximum is $\frac{\sqrt{m}\,D}{n}$. Determine the value of m + n, if m and n are coprime numbers.



WO0332

18. Two microwave coherent point sources emitting waves of wavelength λ are placed at 5λ distance apart. The interference is being observed on a flat non-reflecting surface along a line passing through one source, in a direction perpendicular to the line joining the two sources (refer figure). Considering λ as 4 mm, calculate the positions of maxima and draw shape of interference pattern. Take initial phase difference between the two sources to be zero.



WO0011

19. In fresnel's biprism experiment the distance between the source and the screen is 1m and that between the source and biprism is 10 cm. The wavelength of light used is 6000Å. The fringe width obtained is 0.03 cm and the refracting angle of biprism is 1°. Then calculate the refractive index of the material of biprism.

WO0333

20. When a mica sheet of thickness 7 microns and $\mu = 1.6$ is placed in the path of one of interfering beams in the biprism experiment then the central fringe gets at the position of seventh bright fringe. What is the wavelength of light used ?

WO0334

21. Light of wavelength 6000Å is incident on a thin glass plate of refractive index 1.5 such that angle of refraction into the plate is 60°. Calculate the smallest thickness of plate which will make it appear dark by reflection.

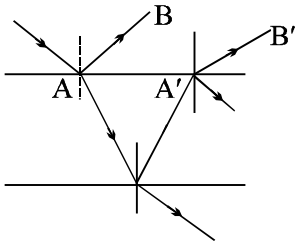
WO0335

22. A long narrow horizontal slit lies 1mm above a plane mirror as in Lloyd's mirror. The interference pattern produced by the slit and its image is viewed on a screen distant 1m from the slit. The wavelength of light is 600nm. Find the distance of first maximum above the mirror.

WO0010

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23. A ray of light of intensity I is incident on a parallel glass-slab at a point A as shown in figure. It undergoes partial reflection and refraction. At each reflection 20% of incident energy is reflected. The rays AB and A′ B′ undergo interference. Find the ratio I_{max}/I_{min}. [Neglect the absorption of light]



WO0013

C. Diffraction

24. Red light of wavelength 6500Å from a distant source falls on a slit 0.50 mm wide. What is the distance between the first two dark bands on each side of the central bright of the diffraction pattern observed on a screen placed 1.8 m. from the slit.

WO0336

25. Light of wavelength 560 nm goes through a pinhole of diameter 0.20 mm and falls on a wall at a distance of 2.00 m. What will be the radius of the central bright spot formed on the wall ?

WO0337

26. Call the intensity at the center of the central Fraunhofer diffraction maximum of a single slit I_0 and the intensity at some other point in the pattern I. Obtain the ratio I/I_0 for a point on the screen that is 3/4 wavelength farther from one edge of the slit than the other.

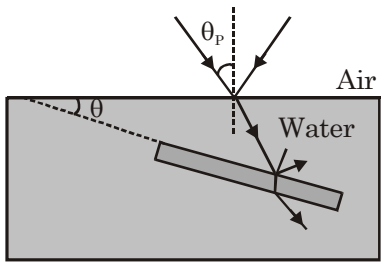
WO0179

27. Your biology textbook tells you that a certain structure in the cell has a size of 0.61 μm. You would like to see this structure using orange light with a wavelength of 600 nm. What minimum numerical aperture must you microscope objective have?

WO0182

D. Polarization

28. Light in air strikes a water surface $\left(n = \frac{4}{3}\right)$ at the polarizing angle. The part of the beam refracted into the water strikes a submerged slab of material with refractive index n as shown in figure. The light reflected from the upper surface of the slab iscompletely polarized. (a) Find the angle θ between the water surface and the surface of the slab as a function of n. (b) Identify the maximum imaginable value of θ and describe the physical situation to which it corresponds. (c) Identify the minimum imaginable value of θ and describe the physical situation to which it corresponds.

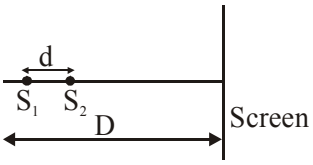


WO0178

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EXERCISE-(JM)

1. Two coherent point sources S_1 and S_2 are separated by a small distance 'd' as shown. The fringes obtained on the screen will be : **[JEE-Mains 2013]**



(1) points
(2) straight lines
(3) semicircles
(4) concentric circles

WO0121

2. During the propagation of electromagnetic waves in a medium : **[JEE-Mains 2014]**
(1) Electric energy density is equal to the magnetic energy density
(2) Both electric magnetic energy densities are zero
(3) Electric energy density is double of the magnetic energy density
(4) Electric energy density is half of the magnetic energy density.

WO0123

3. A red LED emits light at 0.1 watt uniformly around it. The amplitude of the electric field of the light at a distance of 1 m from the diode is:- **[JEE-Mains 2015]**
(1) 5.48 V/m (2) 7.75 V/m (3) 1.73 V/m (4) 2.45 V/m

WO0125

4. Assuming human pupil to have a radius of 0.25cm and a comfortable viewing distance of 25cm, the minimum separation between two objects that human eye can resolve at 500 nm wavelength is :- **[JEE-Mains 2015]**
(1) 100 µm (2) 300 µm (3) 1 µm (4) 30 µm

WO0126

5. On a hot summer night, the refractive index of air is smallest near the ground and increases with height from the ground. When a light beam is directed horizontally, the Huygens' principle leads us to conclude that as it travels, the light beam : **[JEE-Mains 2015]**
(1) bends downwards
(2) bends upwards
(3) becomes narrower
(4) goes horizontally without any deflection

WO0127

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6. The box of a pin hole camera, of length L, has a hole of radius a. It is assumed that when the hole is illuminated by a parallel beam of light of wavelength λ the spread of the spot (obtained on the opposite wall of the camera) is the sum of its geometrical spread and the spread due to diffraction. The spot would then have its minimum size (say b_{min}) when :- **[JEE-Mains 2016]**

(1) $a=\frac{\lambda^2}{L}$ and $b_{min}=\sqrt{4\lambda L}$ (2) $a=\frac{\lambda^2}{L}$ and $b_{min}=\left(\frac{2\lambda^2}{L}\right)$

(3) $a=\sqrt{\lambda L}$ and $b_{min}=\left(\frac{2\lambda^2}{L}\right)$ (4) $a=\sqrt{\lambda L}$ and $b_{min}=\sqrt{4\lambda L}$

WO0128

7. In a Young's double slit experiment, slits are separated by 0.5 mm, and the screen is placed 150 cm away. A beam of light consisting of two wavelengths, 650 nm and 520 nm, is used to obtain interference fringes on the screen. The least distance from the common central maximum to the point where the bright fringes due to both the wavelengths coincide is :- **[JEE-Main 2017]**

(1) 9.75 mm (2) 15.6 mm (3) 1.56 mm (4) 7.8 mm

WO0130

8. An observer is moving with half the speed of light towards a stationary microwave source emitting waves at frequency 10 GHz. What is the frequency of the microwave measured by the observer? (speed of light = 3×10^8 ms^{-1}) **[JEE Main - 2017]**

(1) 17.3 GHz (2) 15.3 GHz (3) 10.1 GHz (4) 12.1 GHz

SW0086

9. The angular width of the central maximum in a single slit diffraction pattern is 60°. The width of the slit is 1 μm. The slit is illuminated by monochromatic plane waves. If another slit of same width is made near it, Young's fringes can be observed on a screen placed at a distance 50 cm from the slits. If the observed fringe width is 1 cm, what is slit separation distance? (i.e. distance between the centres of each slit.) **[JEE-Main 2018]**

(1) 50 μm (2) 75 μm (3) 100 μm (4) 25 μm

WO0131

10. An EM wave from air enters a medium. The electric fields are $\vec{E}_1=E_{01}\hat{x}\cos\left[2\pi\nu\left(\frac{z}{c}-t\right)\right]$ in air and $\vec{E}_2=E_{02}\hat{x}\cos\left[k(2z-ct)\right]$ in medium, where the wave number k and frequency v refer to their values in air. The medium is non-magnetic. If $\in_{r_1}$ and $\in_{r_2}$ refer to relative permittivity of air and medium respectively, which of the following options is correct ? **[JEE-Main 2018]**

(1) $\frac{\in_{r_1}}{\in_{r_2}}=2$ (2) $\frac{\in_{r_1}}{\in_{r_2}}=\frac{1}{4}$ (3) $\frac{\in_{r_1}}{\in_{r_2}}=\frac{1}{2}$ (4) $\frac{\in_{r_1}}{\in_{r_2}}=4$

WO0132

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

11. If the magnetic field of a plane electromagnetic wave is given by (The speed of light = 3×10^8/m/s)

$B = 100 \times 10^{-6} \sin\left[2\pi\times 2\times 10^{15}\left(t-\frac{x}{c}\right)\right]$ then the maximum electric field associated with it is :

[JEE-Main 2019_Jan]

(1) 4×10^4 N/C (2) 4.5×10^4 N/C (3) 6×10^4 N/C (4) 3×10^4 N/C

WO0227

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12. The magnetic field of an electromagnetic wave is given by :-

$$\vec{B}=1.6\times10^{-6}\cos\left(2\times10^{7}z+6\times10^{15}t\right)\left(2\hat{i}+\hat{j}\right)\frac{Wb}{m^2}$$

The associated electric field will be :- **[JEE-Main 2019_April]**

(1) $\vec{E}=4.8\times10^{2}\cos\left(2\times10^{7}z+6\times10^{15}t\right)\left(\hat{i}-2\hat{j}\right)\frac{V}{m}$

(2) $\vec{E}=4.8\times10^{2}\cos\left(2\times10^{7}z-6\times10^{15}t\right)\left(2\hat{i}+\hat{j}\right)\frac{V}{m}$

(3) $\vec{E}=4.8\times10^{2}\cos\left(2\times10^{7}z-6\times10^{15}t\right)\left(-2\hat{j}+\hat{i}\right)\frac{V}{m}$

(4) $\vec{E}=4.8\times10^{2}\cos\left(2\times10^{7}z+6\times10^{15}t\right)\left(-\hat{i}+2\hat{j}\right)\frac{V}{m}$

WO0229

13. In a Young's double slit experiment, the slits are placed 0.320 mm apart. Light of wavelength $\lambda = 500$ nm is incident on the slits. The total number of bright fringes that are observed in the angular range $-30° \le \theta \le 30°$is: **[JEE Main-2019_Jan]**

(1) 320 (2) 641 (3) 321 (4) 640

WO0151

14. In a double-slit experiment, green light (5303 Å) falls on a double slit having a separation of 19.44 μm and a width of 4.05 μm. The number of bright fringes between the first and the second diffraction minima is :- **[JEE Main-2019_Jan]**

(1) 09 (2) 10 (3) 04 (4) 05

WO0152

15. A light wave is incident normally on a glass slab of refractive index 1.5. If 4% of light gets reflected and the amplitude of the electric field of the incident light is 30V/m, then the amplitude of the electric field for the wave propogating in the glass medium will be: **[JEE Main-2019_Jan]**

(1) 10 V/m (2) 24 V/m (3) 30 V/m (4) 6 V/m

WO0153

16. Calculate the limit of resolution of a telescope objective having a diameter of 200 cm, if it has to detect light of wavelength 500 nm coming from a star :- **[JEE Main-2019_April]**

(1) 305×10^{-9} radian (2) 152.5×10^{-9} radian (3) 610×10^{-9} radian (4) 457.5×10^{-9} radian

WO0156

17. Diameter of the objective lens of a telescope is 250 cm. For light of wavelength 600nm. coming from a distant object, the limit of resolution of the telescope is close to :- **[JEE Main-2019_April]**

(1) 1.5×10^{-7} rad (2) 2.0×10^{-7} rad (3) 3.0×10^{-7} rad (4) 4.5×10^{-7} rad

WO0157

18. The value of numerical aperature of the objective lens of a microscope is 1.25. If light of wavelength 5000 Å is used, the minimum separation between two points, to be seen as distinct, will be :

[JEE Main-2019_April]

(1) 0.24 μm (2) 0.48 μm (3) 0.12 μm (4) 0.38 μm

WO0158

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19. The electric fields of two plane electromagnetic plane waves in vacuum are given by

$\vec{E}_1 = E_0\hat{j}\cos(\omega t - kx)$ and

$\vec{E}_2 = E_0\hat{k}\cos(\omega t - ky)$

At t = 0, a particle of charge q is at origin with a velocity $\vec{v} = 0.8c\hat{j}$ (c is the speed of light in vacuum). The instantaneous force experienced by the particle is : **[JEE-Main 2020_Jan]**

(1) $E_0q(-0.8\hat{i}+\hat{j}+\hat{k})$ (2) $E_0q(0.8\hat{i}-\hat{j}+0.4\hat{k})$ (3) $E_0q(0.8\hat{i}+\hat{j}+0.2\hat{k})$ (4) $E_0q(0.4\hat{i}-3\hat{j}+0.8\hat{k})$

WO0233

20. A plane electromagnetic wave is propagating along the direction $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$, with its polarization along the direction $\hat{k}$. The correct form of the magnetic field of the wave would be (here B_0 is an appropriate constant) : **[JEE Main-2020_Jan]**

(1) $B_0\frac{\hat{i}-\hat{j}}{\sqrt{2}}\cos\left(\omega t - k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$ (2) $B_0\frac{\hat{i}+\hat{j}}{\sqrt{2}}\cos\left(\omega t - k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

(3) $B_0\hat{k}\cos\left(\omega t - k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$ (4) $B_0\frac{\hat{j}-\hat{i}}{\sqrt{2}}\cos\left(\omega t + k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

WO0160

21. A plane electromagnetic wave, has frequency of 2.0×10^{10} Hz and its energy density is 1.02×10^{-8} J/ m^3 in vacuum. The amplitude of the magnetic field of the wave is close to($\frac{1}{4\pi\varepsilon_0} = 9\times10^9 \frac{Nm^2}{C^2}$ and speed of light = 3×10^8 ms^{-1}) : **[JEE-Main 2020_Sep]**

(1) 180 nT (2) 160 nT (3) 150 nT (4) 190 nT

WO0234

22. In a plane electromagnetic wave, the directions of electric field and magnetic field are represented by $\hat{k}$ and $2\hat{i}-2\hat{j}$, respectively. What is the unit vector along direction of propagation of the wave.

[JEE-Main 2020_Sep]

(1) $\frac{1}{\sqrt{2}}(\hat{i}+\hat{j})$ (2) $\frac{1}{\sqrt{5}}(\hat{i}+2\hat{j})$ (3) $\frac{1}{\sqrt{5}}(2\hat{i}+\hat{j})$ (4) $\frac{1}{\sqrt{2}}(\hat{j}+\hat{k})$

WO0235

23. The correct match between the entries in column I and column II are :

	I	II
	Radiation	**Wavelength**
(a)	Microwave	(i) 100m
(b)	Gamma rays	(ii) 10^{-15} m
(c)	A.M. radio waves	(iii) 10^{-10} m
(d)	X-rays	(iv) 10^{-3} m

[JEE-Main 2020_Sep]

(1) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii) (2) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

(3) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv) (4) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

WO0239

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound\Wave-EM\Wave & WaveOp\Eng\03_EM-Wave & WO\01_Th+Ex.p65

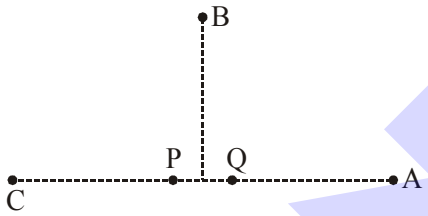
24. A beam of plane polarised light of large cross sectional area and uniform intensity of 3.3 Wm^{-2} falls normally on a polariser (cross sectional area 3×10^{-4} m^2) which rotates about its axis with an angular speed of 31.4 rad/s. The energy of light passing through the polariser per revolution, is close to :

[JEE Main-2020_Sep]

(1) 1.0×10^{-5} J (2) 5.0×10^{-4} J (3) 1.0×10^{-4} J (4) 1.5×10^{-4} J

WO0162

25. In the figure below, P and Q are two equally intense coherent sources emitting radiation of wavelength 20 m. The separation between P and Q is 5 m and the phase of P is ahead of that of Q by 90°. A, B and C are three distinct points of observation, each equidistant from the midpoint of PQ. The intensities of radiation at A, B, C will be in the ratio: **[JEE Main-2020_Sep]**



(1) 0 : 1 : 2 (2) 4 : 1 : 0 (3) 0 : 1 : 4 (4) 2 : 1 : 0

WO0164

26. A radiation is emitted by 1000 W bulb and it generates an electric field and magnetic field at P, placed at a distance of 2 m. The efficiency of the bulb is 1.25%. The value of peak electric field at P is $x \times 10^{-1}$ V/m. Value of x is_. (Rounded-off to the nearest integer)

[Take $\varepsilon_0 = 8.85 \times 10^{-12}$ C^2N^{-1} m^{-2}, $c = 3 \times 10^8$ ms^{-1}] **[JEE Main-2021_Feb]**

WO0241

27. Consider the diffraction pattern obtained from the sunlight incident on a pinhole of diameter 0.1 μm. If the diameter of the pinhole is slightly increased, it will affect the diffraction pattern such thtat :

(1) its size decreases, and intensity decreases
(2) its size increases, and intensity increases
(3) its size increases, but intensity decreases
(4) its size decreases, but intensity increases

[JEE Main-2021_Feb]

WO0338

28. A plane electromagnetic wave propagating along y-direction can have the following pair of electric field $(\vec{E})$ and magnetic field $(\vec{B})$ components. **[JEE Main-2021_March]**

(1) E_y, B_y or E_z, B_z (2) E_y, B_x or E_x, B_y (3) E_x, B_z or E_z, B_x (4) E_x, B_y or E_y, B_x

WO0244

29. A plane electromagnetic wave of frequency 500 MHz is travelling in vacuum along y-direction. At a particular point in space and time, $\vec{B} = 8.0 \times 10^{-8}\,\hat{z}T$. The value of electric field at this point is : (speed of light = 3×10^8 ms^{-1}) $\hat{x}$, $\hat{y}$, $\hat{z}$ are unit vectors along x, y and z direction.

[JEE Main-2021_March]

(1) $-24\,\hat{x}$ V / m (2) $2.6\,\hat{x}$ V / m (3) $24\,\hat{x}$ V / m (4) $-2.6\,\hat{y}$ V / m

WO0339

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound\Wave-EM Wave & Wave Optics\Eng\03_EM-Wave & WO\01_Th+Ex.p65

30. A galaxy is moving away from the earth at a speed of 286 kms^{-1}. The shift in the wavelength of a red line at 630 nm is $x \times 10^{-10}$ m. The value of x, to the nearest integer, is_____.
[Take the value of speed of light c, as 3×10^8 ms^{-1}] **[JEE Main-2021_March]**
WO0340

31. The electric field in a plane electromagnetic wave is given by

$$\vec{E}=200\cos\left[\left(\frac{0.5\times10^3}{m}\right)x-\left(1.5\times10^{11}\frac{rad}{s}\times t\right)\right]\frac{V}{m}\hat{j}$$

If this wave falls normally on a perfectly reflecting surface having an area of 100 cm^2. If the radiation pressure exerted by the E.M. wave on the surface during a 10 minute exposure is $\frac{x}{10^9}\frac{N}{m^2}$. Find the value of x. **[JEE Main-2021_Aug]**
WO0341

32. White light is passed through a double slit and interference is observed on a screen 1.5 m away. The separation between the slits is 0.3 mm. The first violet and red fringes are formed 2.0 mm and 3.5 mm away from the central white fringes. The difference in wavelengths of red and voilet light is ..... nm. **[JEE Main-2021_Aug]**
WO0342

33. The width of one of the two slits in a Young's double slit experiment is three times the other slit. If the amplitude of the light coming from a slit is proportional to the slit-width, the ratio of minimum to maximum intensity in the interference pattern is x : 4 where x is _____.
[JEE Main-2021_Sep]
WO0343

34. If electric field intensity of a uniform plane electro magnetic wave is given as

$$E = -301.6\sin(kz-\omega t)\,\hat{a}_x + 452.4\sin(kz-\omega t)\,\hat{a}_y\frac{V}{m}$$

Then, magnetic intensity H of this wave in Am^{-1} will be:'
[Given: Speed of light in vacuum c = 3×10^8 ms^{-1}, permeability of vacuum $\mu_0 = 4\pi \times 10^{-7}$ NA^{-2}]
[JEE Main-2022_June]

(A) $+0.8\sin(kz-\omega t)\hat{a}_y + 0.8\sin(kz-\omega t)\hat{a}_x$

(B) $+1.0\times10^{-6}\sin(kz-\omega t)\hat{a}_y + 1.5\times10^{-6}(kz-\omega t)\hat{a}_x$

(C) $-0.8\sin(kz-\omega t)\hat{a}_y - 1.2\sin(kz-\omega t)\hat{a}_x$

(D) $-1.0\times10^{-6}\sin(kz-\omega t)\hat{a}_y - 1.5\times10^{-6}\sin(kz-\omega t)\hat{a}_x$

WO0344

35. The displacement current of 4.425 μA is developed in the space between the plates of parallel plate capacitor when voltage is changing at a rate of 10^6 Vs^{-1}. The area of each plate of the capacitor is $40cm^2$. The distance between each plate of the capacitor is $x \times 10^{-3}$m. The value of x is,
(Permittivity of free space, $E_0 = 8.85 \times 10^{-12}$ C^2 N^{-1} m^{-2}) **[JEE Main-2022_June]**
WO0345

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36. A light whose electric field vectors are completely removed by using a good Polaroid, allowed to incident on the surface of the prism at Brewster's angle. Choose the most suitable option for the phenomenon related to the prism. **[JEE Main-2022_June]**

(A) Reflected and refracted rays will be perpendicular to each other

(B) Wave will propagate along the surface of prism

(C) No refraction, and there will be total reflection of light.

(D) No reflection and there will be total transmission of light.

WO0346

37. In a double slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the plane of slits. If the screen is moved by 5×10^{-2} m towards the slits, the change in fringe width is 3×10^{-3} cm. If the distance between the slits is 1 mm, then the wavelength of the light will be _____nm. **[JEE Main-2022_June]**

WO0347

38. Sodium light of wavelengths 650 nm and 655 nm is used to study diffraction at a single slit of aperture 0.5 mm. The distance between the slit and the screen is 2.0 m. The separation between the positions of the first maxima of diffraction pattern obtained in the two cases is _____ $\times 10^{-5}$ m. **[JEE Main-2022_June]**

WO0348

39. Identify the correct statements from the following descriptions of various properties of electromagnetic waves.

A. In a plane electromagnetic wave electric field and magnetic field must be perpendicular to each other and direction of propagation of wave should be along electric field or magnetic field.

B. The energy in electromagnetic wave is divided equally between electric and magnetic fields.

C. Both electric field and magnetic field are parallel to each other and perpendicular to the direction of propagation of wave.

D. The electric field, magnetic field and direction of propagation of wave must be perpendicular to each other.

E. The ratio of amplitude of magnetic field to the amplitude of electric field is equal to speed of light.

Choose the most appropriate answer from the options given below: **[JEE Main-2022_July]**

(A) D only (B) B and D only (C) B, C and E only (D) A, B and E only

WO0349

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EXERCISE-(JA)

1. In the Young's double slit experiment using a monochromatic light of wavelength λ, the path difference (in terms of an integer n) corresponding to any point having half the peak intensity is :-

[JEE Advanced 2013]

(A) $(2n+1)\frac{\lambda}{2}$ (B) $(2n+1)\frac{\lambda}{4}$

(C) $(2n+1)\frac{\lambda}{8}$ (D) $(2n+1)\frac{\lambda}{16}$

WO0135

2. A light source, which emits two wavelengths $\lambda_1 = 400$ nm and $\lambda_2 = 600$ nm, is used in a Young's double slit experiment. If recorded fringe widths for λ_1 and λ_2 are β_1 and β_2 and the number of fringes for them within a distance y on one side of the central maximum are m_1 and m_2, respectively, then :- **[JEE Advanced 2014]**

(A) $\beta_2 > \beta_1$

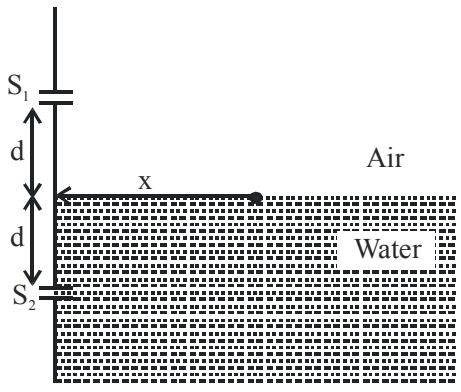
(B) $m_1 > m_2$

(C) From the central maximum, 3rd maximum of λ_2 overlaps with 5th minimum of λ_1

(D) The angular separation of fringes of λ_1 is greater than λ_2

WO0136

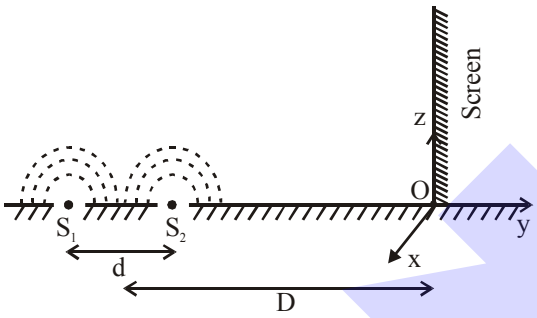
3. A Young's double slit interference arrangement with slits S_1 and S_2 is immersed in water (refractive index = 4/3) as shown in the figure. The positions of maxima on the surface of water are given by $x^2 = p^2m^2\lambda^2 - d^2$, where λ is the wavelength of light in air (refractive index = 1), 2d is the separation between the slits and m is an integer. The value of p is. **[JEE Advanced 2015]**



WO0137

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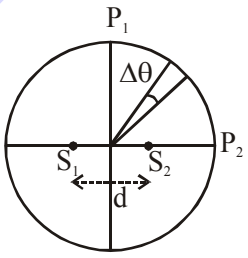
4. While conducting the Young's double slit experiment, a student replaced the two slits with a large opaque plate in the x-y plane containing two small holes that act as two coherent point sources (S_1, S_2) emitting light of wavelength 600 nm. The student mistakenly placed the screen parallel to the x-z plane (for $z > 0$) at a distance D = 3m from the mid-point of S_1S_2, as shown schematically in the figure. The distance between the sources d = 0.6003 mm. The origin O is at the intersection of the screen and the line joining S_1S_2. Which of the following is (are) true of the intensity pattern on the screen ? **[JEE Advanced 2016]**



(A) Hyperbolic bright and dark bands with foci symmetrically placed about O in the x-direction
(B) Semi circular bright and dark bands centered at point O
(C) The region very close to the point O will be dark
(D) Straight bright and dark bands parallel to the x-axis

WO0138

5. Two coherent monochromatic point sources S_1 and S_2 of wavelength λ = 600 nm are placed symmetrically on either side of the center of the circle as shown. The sources are separated by a distance d = 1.8mm. This arrangement produces interference fringes visible as alternate bright and dark spots on the circumference of the circle. The angular separation between two consecutive bright spots is $\Delta\theta$. Which of the following options is/are correct ? **[JEE Advanced 2017]**

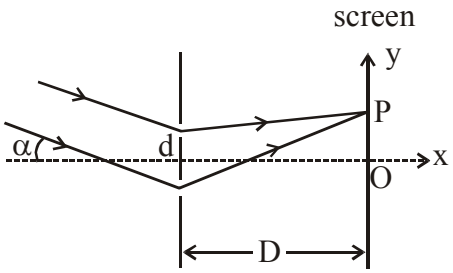


(A) A dark spot will be formed at the point P_2
(B) The angular separation between two consecutive bright spots decreases as we move from P_1 to P_2 along the first quadrant
(C) At P_2 the order of the fringe will be maximum
(D) The total number of fringes produced between P_1 and P_2 in the first quadrant is close to 3000

WO0139

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6. In a Young's double slit experiment, the slit separation d is 0.3 mm and the screen distance D is 1m. A parallel beam of light of wavelength 600nm is incident on the slits at angle α as shown in figure. On the screen, the point O is equidistant from the slits and distance PO is 11.0 mm. Which of the following statement(s) is/are correct ? **[JEE Advanced 2019]**



(1) For $\alpha = \frac{0.36}{\pi}$ degree, there will be destructive interference at point O.

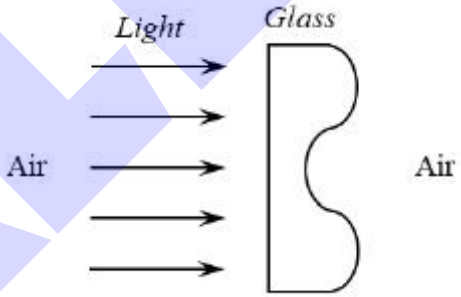
(2) Fringe spacing depends on α

(3) For $\alpha = \frac{0.36}{\pi}$ degree, there will be destructive interference at point P

(4) For $\alpha = 0$, there will be constructive interference at point P.

WO0140

7. A parallel beam of light strikes a piece of transparent glass having cross section as shown in the figure below. Correct shape of the emergent wavefront will be (figures are schematic and not drawn to scale)- **[JEE Advanced 2020]**



(A) 

(B) 

(C) 

(D) 

WO0165

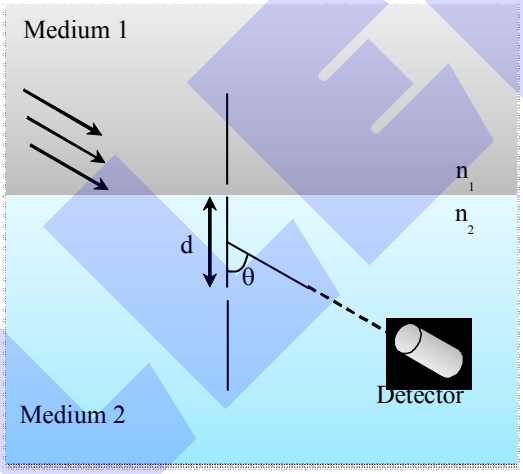
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8. A physical quantity $\vec{S}$ is defined as $\vec{S} = (\vec{E} \times \vec{B}) / \mu_0$, where $\vec{E}$ is electric field, $\vec{B}$ is magnetic field and μ_0 is the permeability of free space. The dimensions of $\vec{S}$ are the same as the dimensions of which of the following quantity (ies) ? **[JEE Advanced 2021]**

(A) $\frac{Energy}{charge \times current}$ (B) $\frac{Force}{Length \times Time}$ (C) $\frac{Energy}{Volume}$ (D) $\frac{Power}{Area}$

WO0350

9. A double slit setup is shown in the figure. One of the slits is in medium 2 of refractive index n_2. The other slit is at the interface of this medium with another medium 1 of refractive index $n_1 (\neq n_2)$. The line joining the slits is perpendicular to the interface and the distance between the slits is d. The slit widths are much smaller than d. A monochromatic parallel beam of light is incident on the slits from medium 1. A detector is placed in medium 2 at a large distance from the slits, and at an angle θ from the line joining them, so that θ equals the angle of refraction of the beam. Consider two approximately parallel rays from the slits received by the detector. **[JEE Advanced 2022]**



Which of the following statement(s) is (are) correct?

(A) The phase difference between the two rays is independent of d.

(B) The two rays interfere constructively at the detector.

(C) The phase difference between the two rays depends on n_1 but is independent of n_2.

(D) The phase difference between the two rays vanishes only for certain values of d and the angle of incidence of the beam, with θ being the corresponding angle of refraction.

WO0351

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ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	D	A	D	B	A	B	C	D	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	C	C	A	C	B	D	D	A	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	C	B	C	C	C	A	D	A	C
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	B	A	A	A	C	C	B	B	A	A
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	B	C	D	A	B	D	A	A	D
Que.	51	52	53	54	55	56				
Ans.	B	C	D	D	B	A				

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	C	D	D	B	C	C	D	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	C	C	B	A	D	D	C	C	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	A,B,C	A,B,C	A,C	A,D	A,C	B,D	B,D	A,C,D	A,B,D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A,C	A,C	B,C,D	A,B,C,D	B,C	A,B,C	A,C,D	A,B,D	B,D	D
Que.	41	42								
Ans.	A	C								

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EXERCISE # S

1. (a) $I_{rms} = V_{rms}\, wC = 6.9$ A (b) Yes (c) $B_0 = \frac{\mu_0}{2\pi}\frac{r}{R^2} i_0$, $B_0 = 1.63 \times 10^{-5}$ T

3. 7 V/cm **4.** $J = \varepsilon_0 \frac{dE}{dt} = \begin{cases} \varepsilon_0 \frac{d}{dt}\left(\frac{r}{2}\frac{dB}{dt}\right) = \frac{\varepsilon_0 r}{2}\ddot{B} \\ \varepsilon_0 \frac{d}{dt}\left(\frac{R^2}{2r}\frac{dB}{dt}\right) = \frac{\varepsilon_0 R^2}{2r}\ddot{B} \end{cases}$ **5.** 52 nT-m

6. (a) $-\hat{j}$ (b) 3.5 m (b) 86 MHz (c) 10.3 nT (e) $\{(10.3\text{ nT})\cos[(1.8\text{ rad/m})\, y + (5.4 \times 10^8\text{ rad/s})t]\}\ \hat{k}$

7. (a) 1.5×10^{-2} m; (b) 1.6×10^{-7} T

8. 15 **10.** (a) a = –14-6, $B_0 = 6.8 \times 10^{-2}$, (b) $-260\hat{i} + 140\hat{j} - 200\hat{k}$ W / m^2

11. $\vec{E} = (5.19 \times 10^2\text{ V/m})\cos[(2.54 \times 10^6\text{ m}^{-1})\, x + (1.47 \times 10^6\text{ m}^{-1})y - (2\pi \times 10^{14}\text{ s}^{-1})t]\,\hat{k}$

12. $\left(n - \frac{1}{48}\right)\lambda = x_1 - x_2$ **13.** 5000Å

14. 1.99×10^{-2} mm **15.** 35.35 cm approximate, 5

16. 8 μm **17.** 7 **18.** 48, 21, $\frac{32}{3}$, $\frac{9}{2}$, 0 m.m.; 

19. 1.573 **20.** 6×10^{-7} meter **21.** 4×10^{-7} m

22. 0.15 mm **23.** 81 : 1 **24.** 4.68 mm

25. 0.68 cm **26.** 0.08 to 0.09 **27.** 0.6

28. (a) $\tan^{-1}\left(\frac{3n}{4}\right) - 37°$; (b) 53° (n = ∞) (c) 0° (n = air)

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EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	4	4	2	4	4	1	4	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	4	2	By NTA 3 By Allen 4	2	1	3	1	3	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	1	4	3	4	137	4	3	1	6
Que.	31	32	33	34	35	36	37	38	39	
Ans.	354	300	1	C	8	D	600	3	B	

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9
Ans.	B	A,B,C	3	B,C	C,D	3	A	B,D	A,B

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